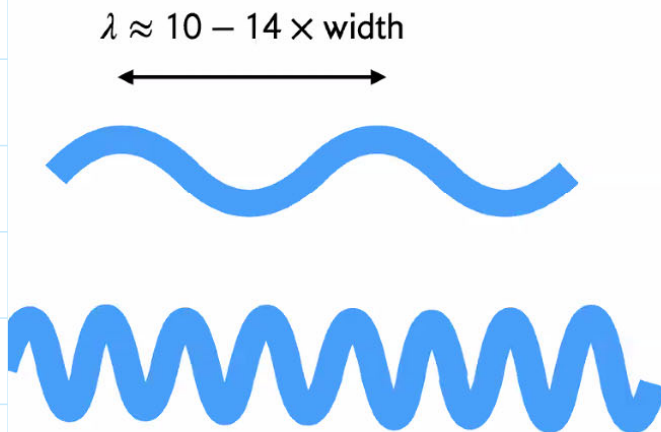


What are already known



- Discharge
- Sediment load

Science

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REPORT

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Meander Wavelength of Alluvial Rivers

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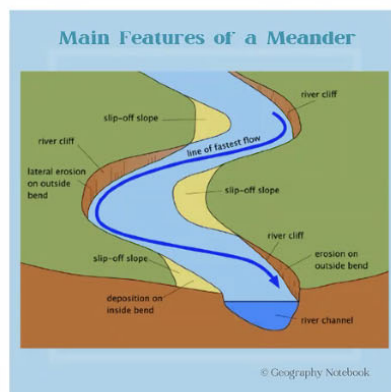
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Abstract

Data on river channel and sediment characteristics were collected at 36 cross sections of stable alluvial river channels in Australia and western United States. These data demonstrate that the meander wavelength of a river is dependent not only on water discharge, but also on the type of sediment load moved through the channel. The meander wavelength of rivers that are transporting a high proportion of their total sediment load as both sand and gravel will be greater than the meander wavelengths of channels of similar discharge which are transporting mainly fine sediment loads.

What are already known

- Erodible banks amplify sinusoidal meander at certain wavelengths (Bend instability theory)



J. Fluid Mech. (1981), vol. 112, pp. 363-377
Printed in Great Britain

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Bend theory of river meanders. Part 1. Linear development

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(Received 31 December 1979 and in revised form 26 June 1981)

Instability of the alternate-bar type in straight channels has long been identified as the cause of fluvial meandering. The condition of inerodible sidewalls, however, does not allow a meandering channel to develop. Herein a stability analysis of a sinuous channel with erodible banks allows for delineation of a 'bend' instability that does not occur in straight channels, and differs from the alternate-bar instability.

In the case of alluvial meanders, the two mechanisms are shown to operate at similar characteristic wavelengths. This provides a rationale for the continuous evolution of alternate bars into true bends such that each bend contains one alternate bar.

The same bend instability applies to incised meanders. A mechanism for incised alternate bars which differs from that for the alluvial case appears to operate at different characteristic wavelengths than that of bend instability. Analysis of data suggests that meandering in supraglacial meltwater streams is primarily due to the alternate bar mechanism, whereas the meandering of rills incised in cohesive material and of caves is likely due to the bend mechanism.

The meander wavelength of incised reaches of meandering streams is often longer than that of adjacent alluvial reaches. An explanation is offered in terms of bend instability.

What are already known

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S. Ikeda, G. Parker and K. Savai

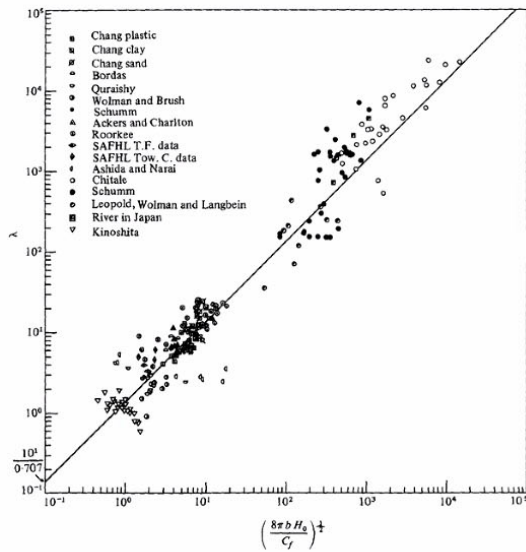


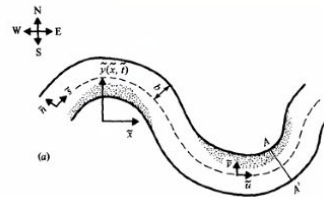
FIGURE 4. Test of equation (30) for alluvial data; the units are metres.

$$\lambda_{\max} \approx \left(\frac{8\pi b H_0}{C_f} \right)^{\frac{1}{2}}$$

b : river half width

H_0 : mean depth

C_f : frictional drag coefficient



Based on St Venant equations

It also predicts downstream migration

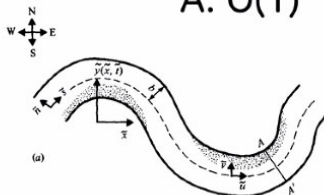
$$\omega_0 = \frac{C_f k^3 (2 + A + F^2)}{k^2 + 4C_f^2}, \quad c_0 = \frac{\omega_0}{k},$$

$k = 2\pi/\lambda$: wavenumber

C_f : frictional drag coefficient

F : Froude number

A : $O(1)$



Based on St Venant equations

Flat plain river

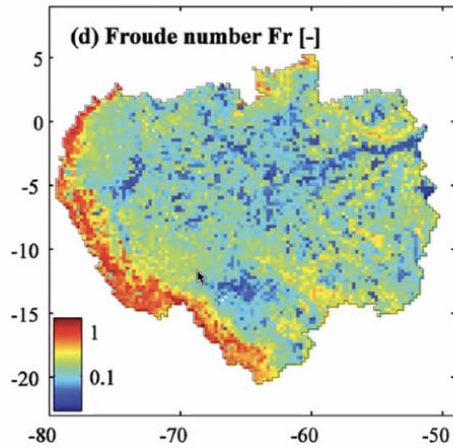
New Approach (vorticity dynamics)

- Alluvial rivers are slow

$$F_r = \frac{U_0}{\sqrt{gH_0}}$$

$$0.1 < F_r < 0.3$$

⇒ Decoupling of fast and slow dynamics



Getirana et al. (2012)

New Approach (vorticity dynamics)

- Flow curvature sustains a background vorticity gradient

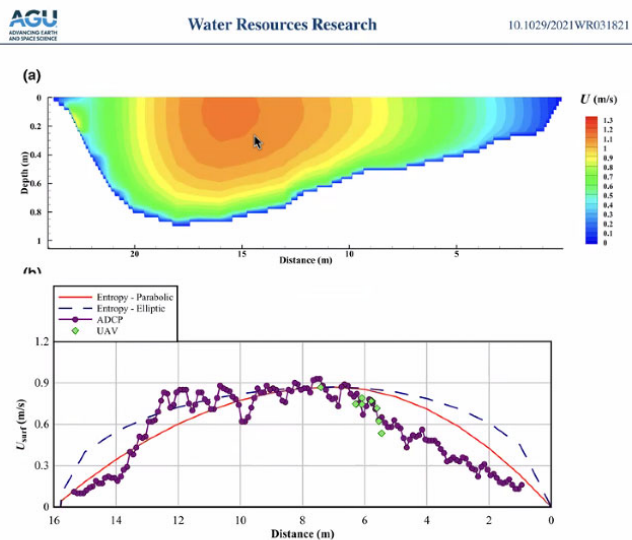
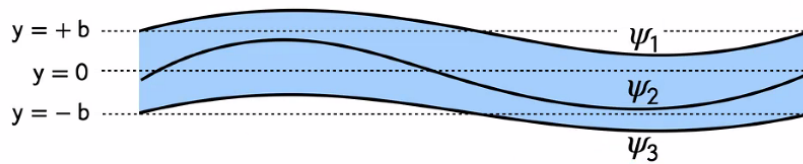


Figure 6. Surface velocity distribution based on parabolic and elliptic distributions, Freiburger Mulde CS3.

Bahmanpouri et al. 2022

New Approach (vorticity dynamics)

- Depth-averaged flow follows vorticity equation



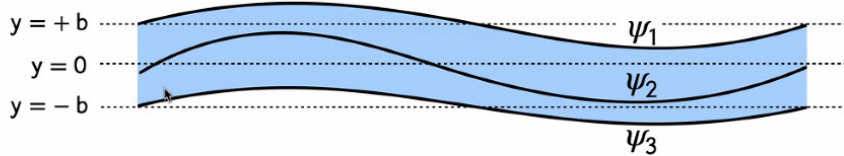
$$\bar{u}(y) = -\frac{\partial \bar{\psi}}{\partial y} = U_0 + \frac{\Delta}{b^2} (b^2 - y^2)$$

$$\begin{aligned}\psi_1(x, t) &= \bar{\psi}(b) + \hat{\psi}_1 \exp[i(kx - \omega t)] \\ \psi_2(x, t) &= \bar{\psi}(0) + \hat{\psi}_2 \exp[i(kx - \omega t)] \\ \psi_3(x, t) &= \bar{\psi}(-b) + \hat{\psi}_3 \exp[i(kx - \omega t)]\end{aligned}$$

$$\begin{aligned}\zeta'_2(x, t) &= \nabla^2 \psi' \\ &\approx \left(\frac{\hat{\psi}_1 + \hat{\psi}_3 - 2\hat{\psi}_2}{b^2} - k^2 \hat{\psi}_2 \right) \exp[i(kx - \omega t)]\end{aligned}$$

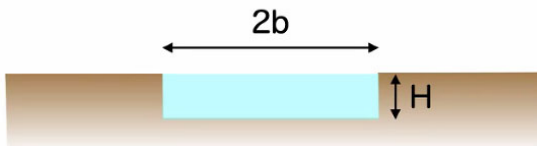
$$\hat{\psi}_1 = \hat{\psi}_3$$

- Depth-averaged flow follows vorticity equation

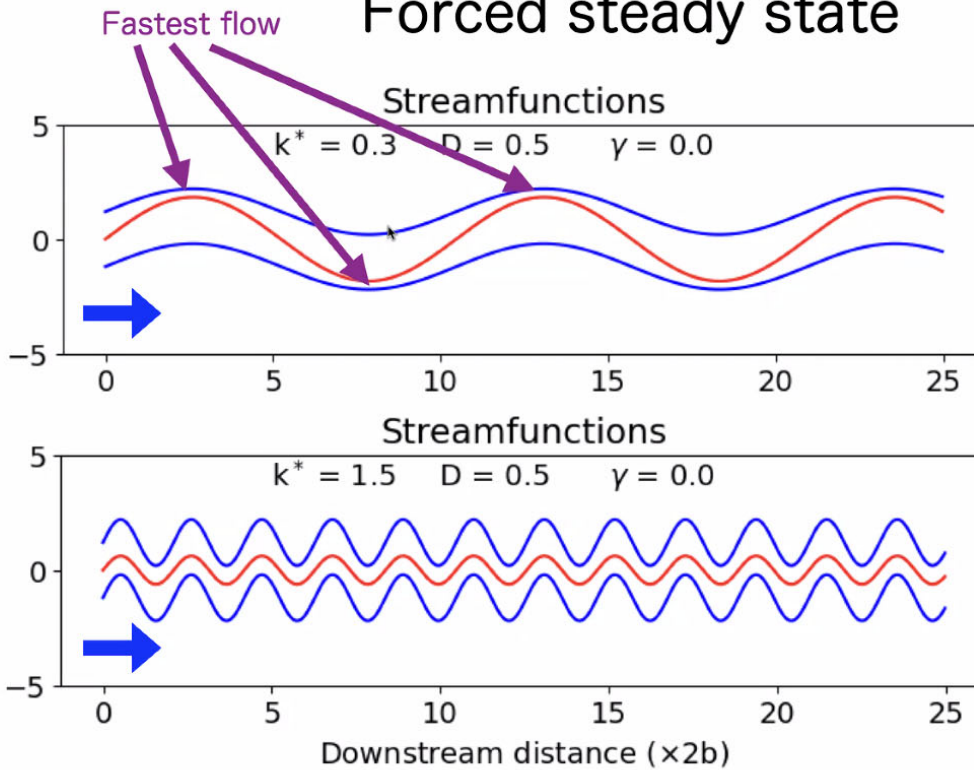


- In steady state, this comes down to

$$\hat{\psi}_2 = f(\hat{\psi}_1)$$



Forced steady state

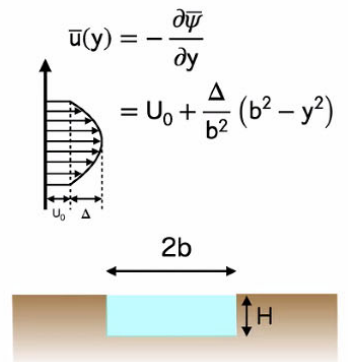


$$\lambda = \frac{2\pi}{k}$$

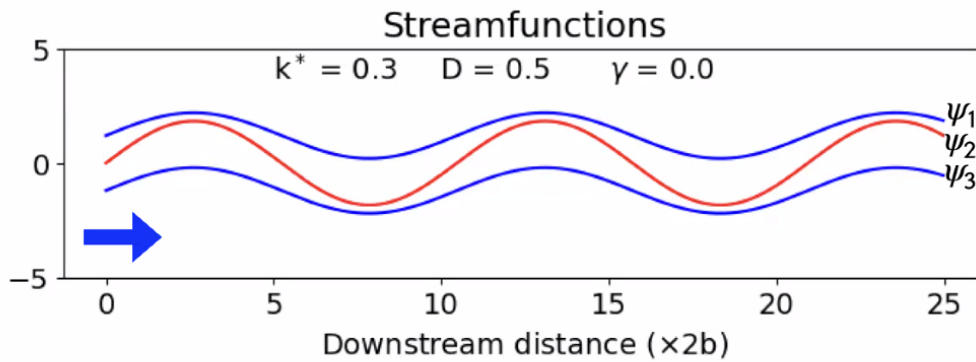
$$k^* = kb$$

$$D = \frac{\Delta}{U_0 + \Delta}$$

$$\gamma = \frac{C_f b}{H}$$

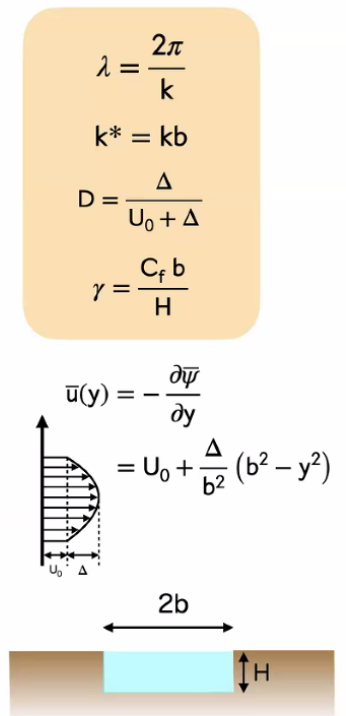


Forced steady state

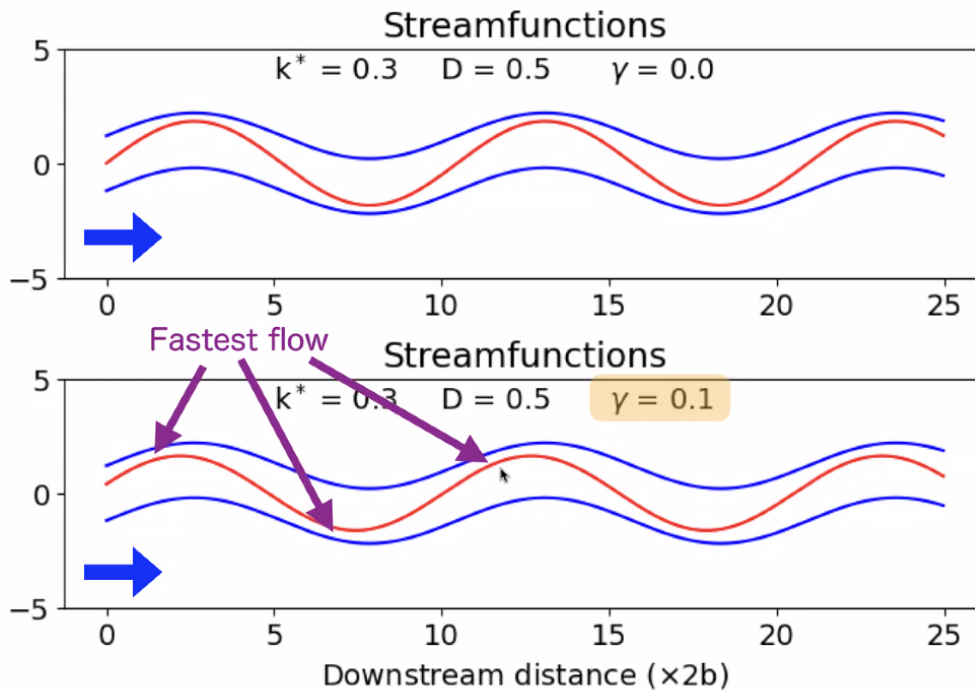


$$|\hat{\psi}_2| > |\hat{\psi}_1| \quad \text{if} \quad k^{*2} < 2D$$

There is no resonance for $k^* > 0$ even in the inviscid limit.



Forced-dissipative steady state

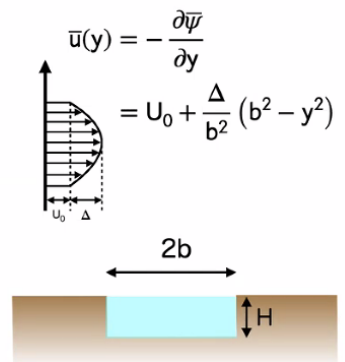


$$\lambda = \frac{2\pi}{k}$$

$$k^* = kb$$

$$D = \frac{\Delta}{U_0 + \Delta}$$

$$\gamma = \frac{C_f b}{H}$$

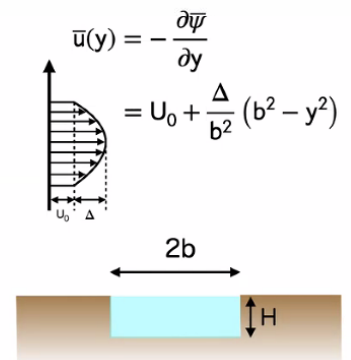
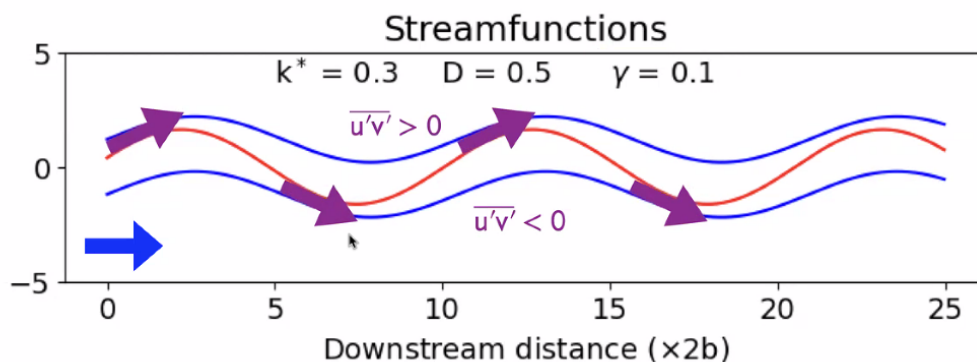


Forced-dissipative steady state

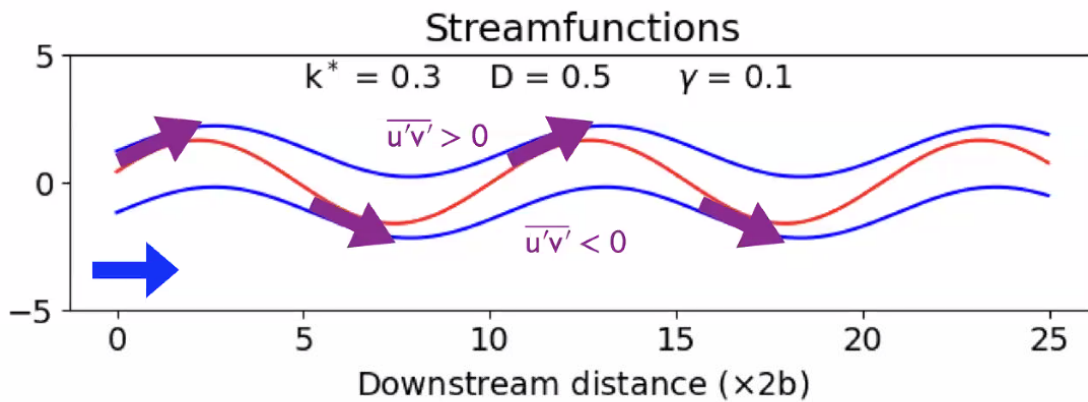
- Momentum is transported out of the channel center to the banks

$$\frac{\partial}{\partial y} \overline{u'v'} \approx -\overline{v'_2 \zeta'_2} = \frac{D\gamma}{2b} \overline{\zeta'^2_2} > 0$$

- Channel flow is dragged by friction at the bottom and sidewalls AND by the pressure drag due to wavy banks



Forced-dissipative steady state

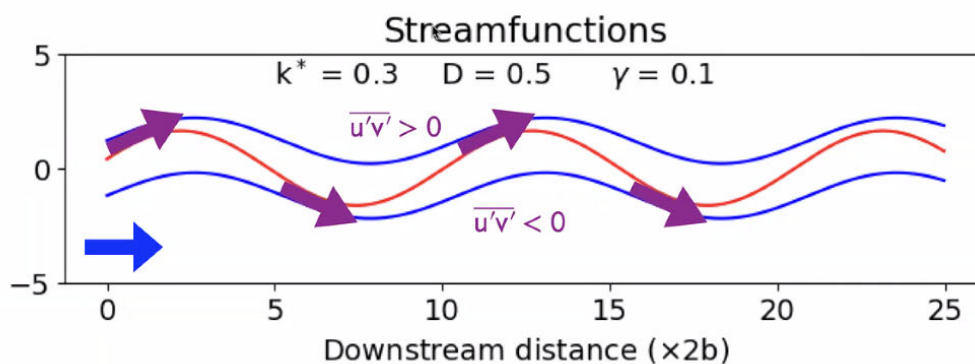


Bank erosion

- This suggests an UPSTREAM propagation of meanders when the banks are erodible

$$\frac{\partial \psi'_1}{\partial t} = \frac{\varepsilon C_f U_0}{b} (\psi'_2 - \psi'_1)$$

$$\mathbf{M}(\omega) \begin{bmatrix} \hat{\psi}'_1 \\ \hat{\psi}'_2 \end{bmatrix} = 0 \Rightarrow \det \mathbf{M} = 0$$

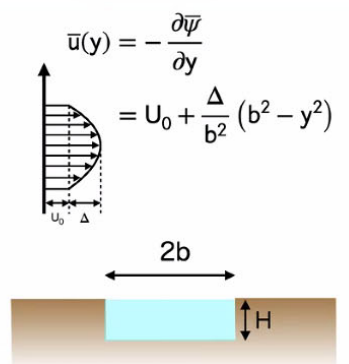


$$\lambda = \frac{2\pi}{k}$$

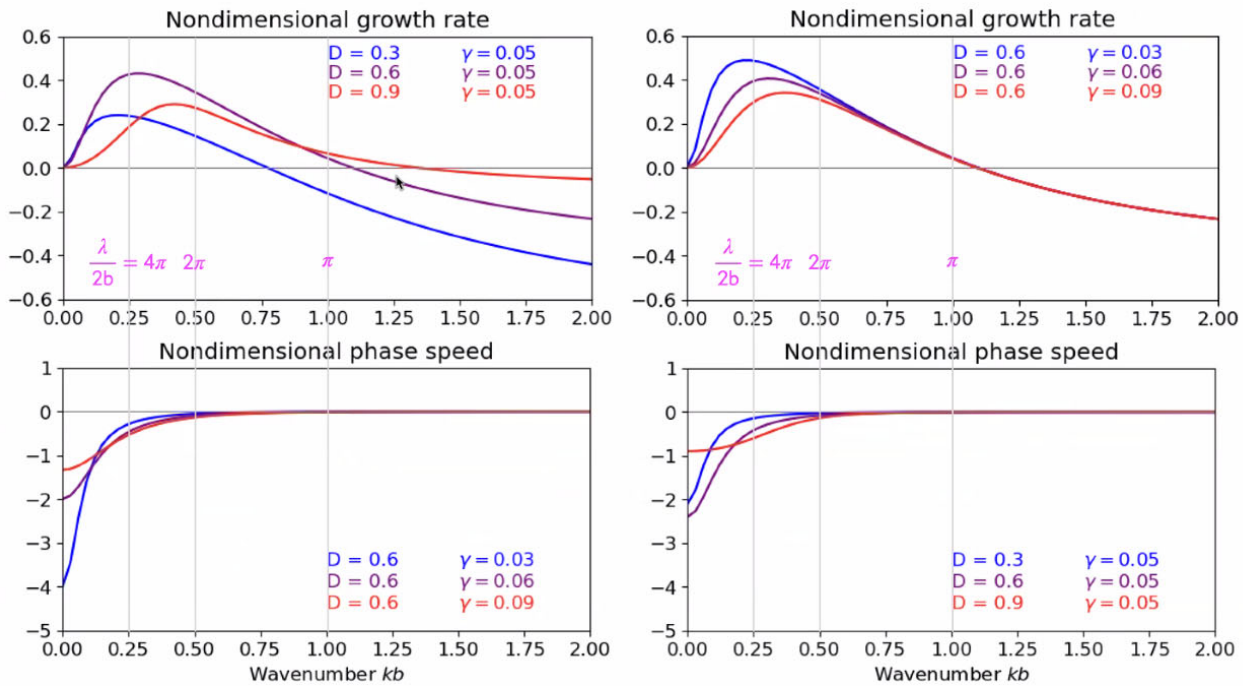
$$k^* = kb$$

$$D = \frac{\Delta}{U_0 + \Delta}$$

$$\gamma = \frac{C_f b}{H}$$



Bend instability revisited



Goals for Rossbypalooza

- Figuring out key differences from previous work
- A numerical model of river flow with erodible banks and bottoms
 - A jet profile with positive vorticity gradient (1D or 2D)
 - Linear fluid dynamics
 - Boundary conditions at the bottom and walls
 - Feedback from sediment transport
- Physical experiments with erodible materials